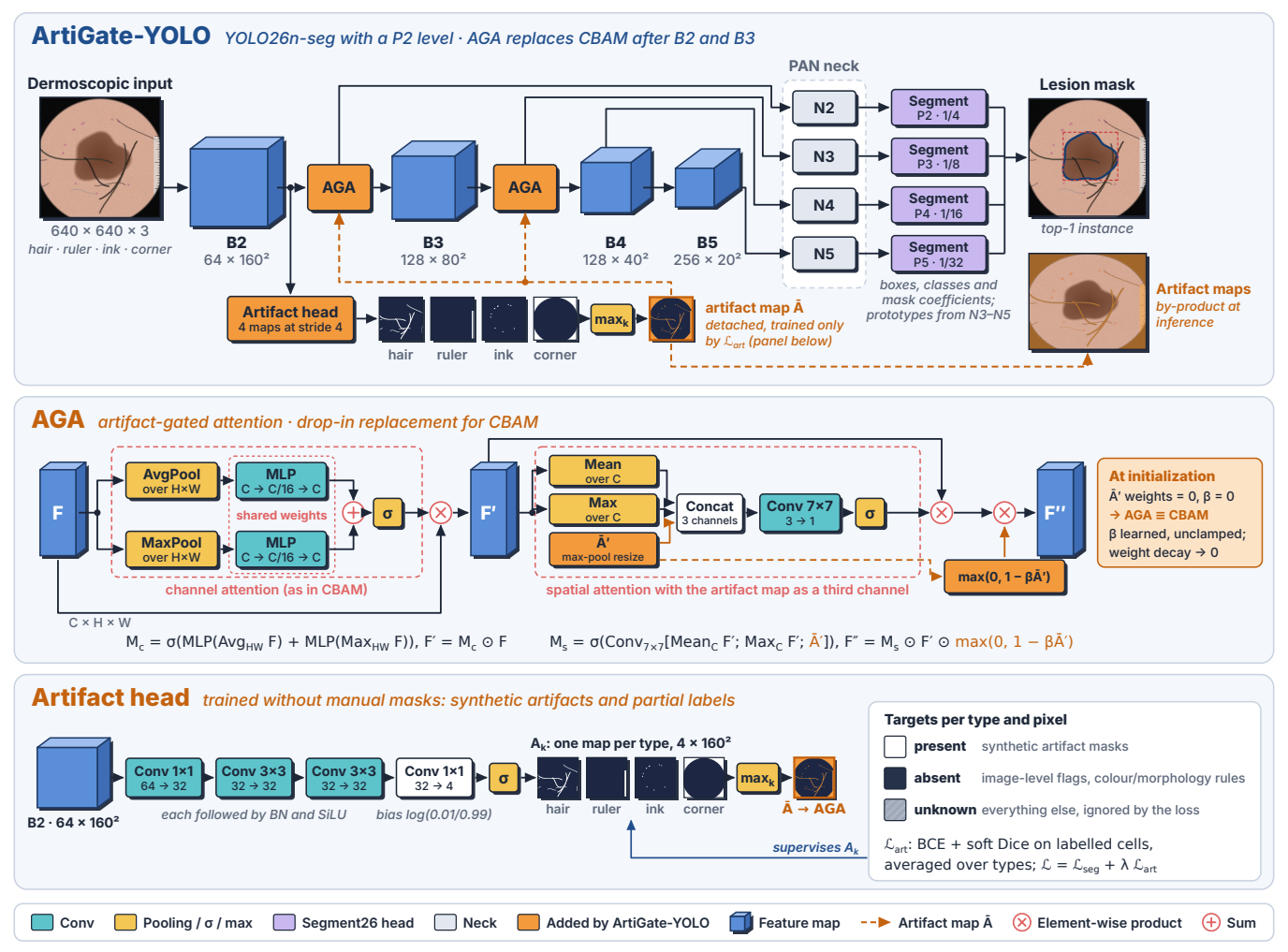

ArtiGate-YOLO
YOLO26n-seg with a P2 level · AGA replaces CBAM after B2 and B3
Dermoscopic input
640 × 640 × 3
hair · ruler · ink · corner
B2
64 × 160²
AGA
B3
128 × 80²
AGA
B4
128 × 40²
B5
256 × 20²
PAN neck
N2
N3
N4
N5
Segment
P2 · 1/4
Segment
P3 · 1/8
Segment
P4 · 1/16
Segment
P5 · 1/32
boxes, classes and mask coefficients; prototypes from N3–N5
Lesion mask
top-1 instance
Artifact maps
by-product at inference
Artifact head
4 maps at stride 4
hair
ruler
ink
corner
max_k
artifact map Ā
detached, trained only by ℒ_art (panel below)
AGA
artifact-gated attention · drop-in replacement for CBAM
F
AvgPool
over H×W
MaxPool
over H×W
MLP
C → C/16 → C
shared weights
MLP
C → C/16 → C
σ
channel attention (as in CBAM)
C × H × W
F′
Mean
over C
Max
over C
Ā′
max-pool resize
Concat
3 channels
Conv 7×7
3 → 1
σ
spatial attention with the artifact map as a third channel
max(0, 1 − βĀ′)
F″
At initialization
Ā′ weights = 0, β = 0
→ AGA ≡ CBAM
β learned, unclamped; weight decay → 0
M_c = σ(MLP(Avg_HW F) + MLP(Max_HW F)), F′ = M_c ⊙ F
M_s = σ(Conv_7×7[Mean_C F′; Max_C F′; Ā′]), F″ = M_s ⊙ F′ ⊙ max(0, 1 − βĀ′)
Artifact head
trained without manual masks: synthetic artifacts and partial labels
B2 · 64 × 160²
Conv 1×1
64 → 32
Conv 3×3
32 → 32
Conv 3×3
32 → 32
Conv 1×1
32 → 4
each followed by BN and SiLU
bias log(0.01/0.99)
σ
A_k: one map per type, 4 × 160²
hair
ruler
ink
corner
max_k
Ā → AGA
supervises A_k
Targets per type and pixel
present synthetic artifact masks
absent image-level flags, colour/morphology rules
unknown everything else, ignored by the loss
ℒ_art: BCE + soft Dice on labelled cells, averaged over types; ℒ = ℒ_seg + λ ℒ_art
Conv
Pooling / σ / max
Segment26 head
Neck
Added by ArtiGate-YOLO
Feature map
Artifact map Ā
Element-wise product
Sum